

Shaft Sinking

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Shaft sinking

- Why shaft is required?
- Why shaft sinking so critical?
- What factors influence selection of shaft?
- What are the hazards of shaft sinking?
- What are the major operations of shaft sinking?

Why shaft is required?

- To provide an entry to mineral deposit at deeper horizon
- To ensure safe travel of men to place of work underground
- To provide passage for ventilating air of required quantity
- For transport of material, mineral or machines
- To provide passage for taking down sources of energy , electrical, pneumatic, or hydraulic and communication system
- To act as sump at pit bottom

Why shaft sinking is a critical operation?

- Shaft is the life line of a mine
- It should remain serviceable throughout the entire life span of a mine
- Stability and integrity of shaft and its fittings are safety critical
- Shaft sinking involves high capital expenditure – has high impact on cost
- Shaft capacity determines the economics of mine

What factors influence selection of shaft?

- Installation from surface or underground
- Vertical or inclined
- Tonnages to be hoisted
- Size of mining equipment to be lowered
- Amount of water to be handled
- Ventilation requirement
- The type of ground through which shaft is to be sunk through
- Position of shaft in relation to ore body

Location of shaft

- Position wrt deposit's geometry
 - Production shaft – at geometrical work load centre of deposit
 - Ore handled in tonne-km should be equal, as far as possible, from all sides
 - For flat deposit, protection pillar around shaft
 - For steeply dipping deposits, on footwall side of ore body



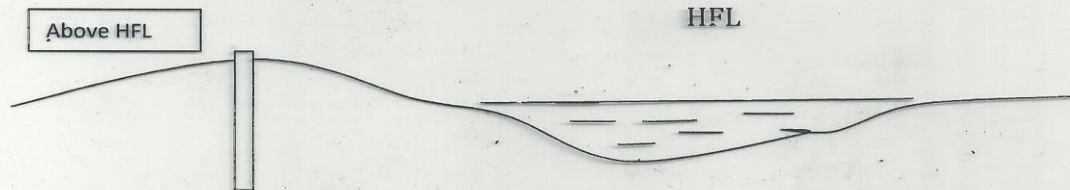
Location of shaft

- Position wrt surface topography
 - Shaft collar at least 5 m above recorded HFL
 - Away from places of public utility and water bodies
 - Within an easy access to infrastructure facilities
 - Enough space around to establish necessary facilities and construction

Location of shaft

- Position wrt geological disturbances, water table and ground condition
 - Type of strata through which shaft will pass through
 - Passing through geologically disturbed strata should be avoided
 - Loose ground, water bearing strata, mud and running sand areas offer difficulties and require special treatment

2. High flood level - above HFL



3. Bearing strength of ground: Adequate to bear the structure of head gear and buildings

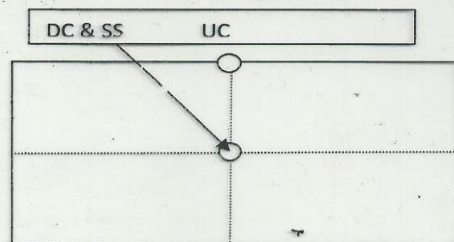
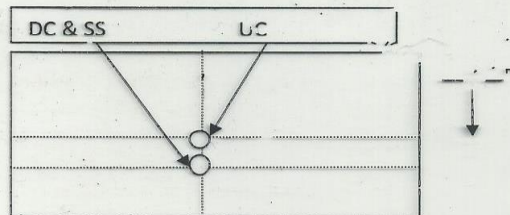
4. Loss of coal in shaft pillar.

5. Land slide in hilly terrain

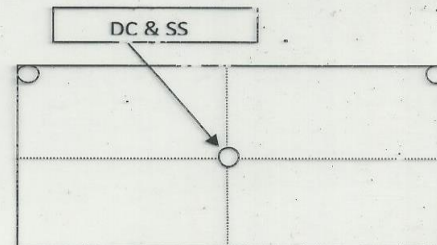
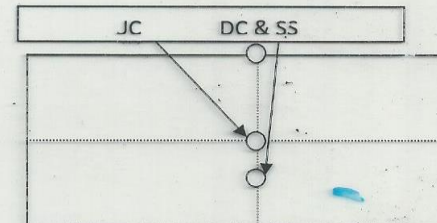
ECONOMIC:

1. T-km Transport Effort is minimum
2. Main and Auxiliary ventilating fans work to their optimum capacity and efficiency.

DIFFERENT LAYOUTS OF LOCATING A SHAFT



(b)



(d)

Hazards of shaft sinking

- Stability of ground or sides
- Noxious or toxic gases
- Temperature and humidity
- Fall from height
- Slip / trip and fall
- Hit by objects / machines / buckets / grabs etc
- Restricted work place
- Blasting projectiles
- Water – drowning

SINKING CYCLE

- Drilling
- Blasting
- Mucking
- Hoisting
- Support or shaft lining
- Auxiliary operations
 - Dewatering
 - Ventilation
 - Lighting
 - Shaft centering

SINKING METHODS

- ORDINARY METHODS

- SPECIAL METHODS

- Piling

- Wooden
 - Steel
 - Concrete

- Caisson

- Pneumatic
 - Without pumping
 - With pumping

- Freezing

- Grouting

- Cement

- Clay

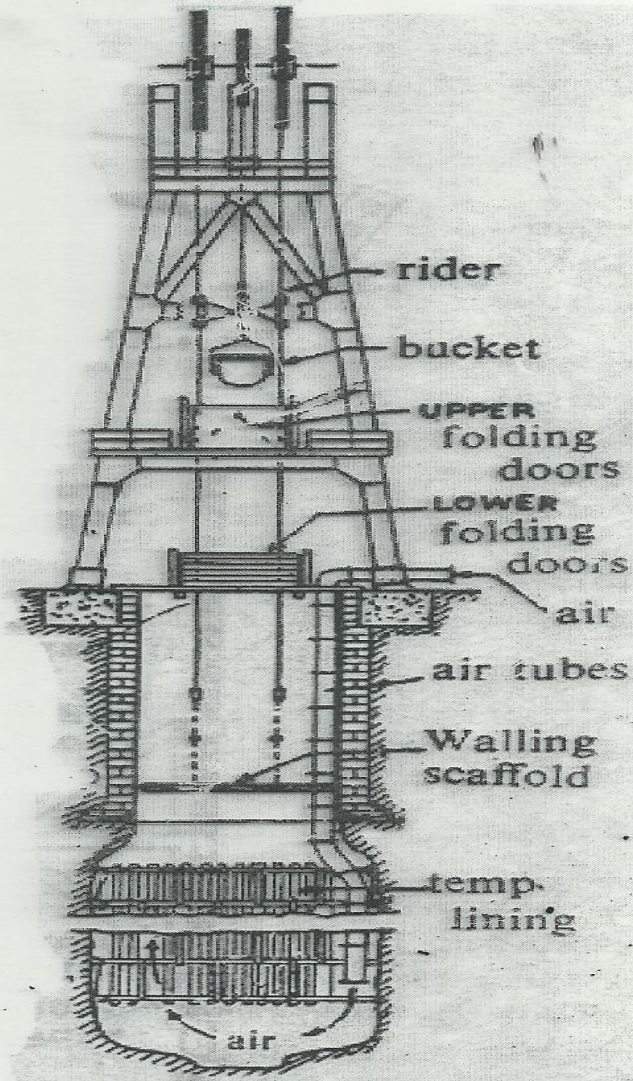
- Chemicals

- Shaft drilling & Boring

General arrangements for shaft sinking

The surface plant and equipment required for sinking is as follows:

1. Steam boilers or diesel engines, unless electric power is available
2. Winding engines and winders fitted with locked coil ropes
3. Steel headgear
4. Double drum winches for walling scaffold, and other winches for lighting cables, shot-firing cables, pump suspension ropes and pump cables
5. Air compressors
6. Fan of nearly 300 cu.m. per minute
7. Generator
8. Folding doors
9. Shaft centering arrangement
10. Signaling arrangements
11. Chutes, buckets and tipping tubs with tramline, etc. should be provided for disposal of debris
12. Workshop
13. Lamp room, first aid room, magazine stores, office, etc.



General Arrangement
of Sinking Shaft.

SEQUENCE OF SINKING OPERATION - DRILLING

- Sinkers – 32-38 mm dia holes
- Shaft jumbos (with no. of drifters) – 40-55mm dia holes
- Hole length – 1.5 to 3 m for sinkers /upto 5m for shaft jumbos
- Drilling pattern – Wedge cut / pyramid cut / step cut
- ☐ • Wedge cut popular in rectangular shaft whereas pyramid cut for circular shaft
- ☐ • Step cut where make of water is high and x-section is large

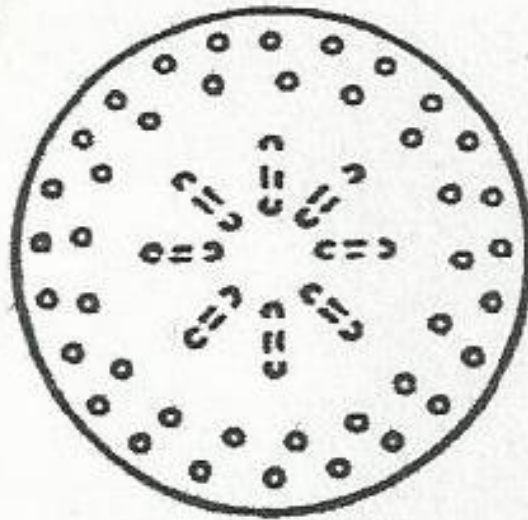
SEQUENCE OF SINKING OPERATION - DRILLING

- No. of holes depend on hole dia, shaft dia, type of strata
- $N = 2.55A + 22$
 - A is cross sectional area in m^2
- For circular shaft, in holes are drilled 3 to 5 concentric circles
- Ratio of holes 1:2:3 or 1:2:3:4:5

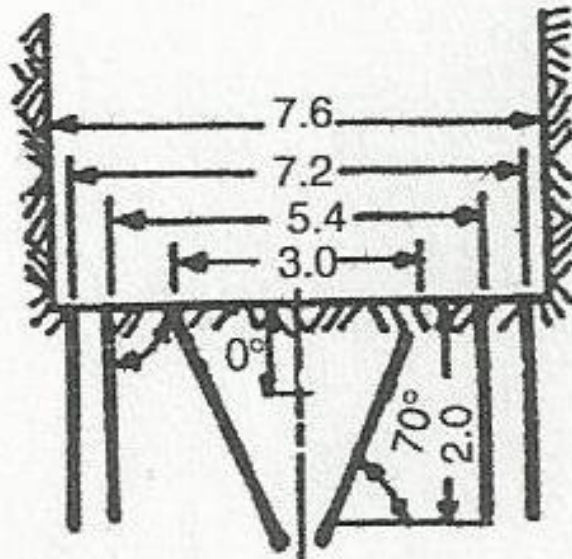
Drill jumbo

- Advantage of drill jumbo:
 - Enhanced Sinking rate and productivity
 - Reduction in cost of sinking
 - Reduced human and machine population
 - Small crews – better management
- Limitation
 - Higher capital cost of rigs and spares
 - Skilled maintenance and operation
 - Kibble winder and opening through working stage – to be matched with jumbo dimension

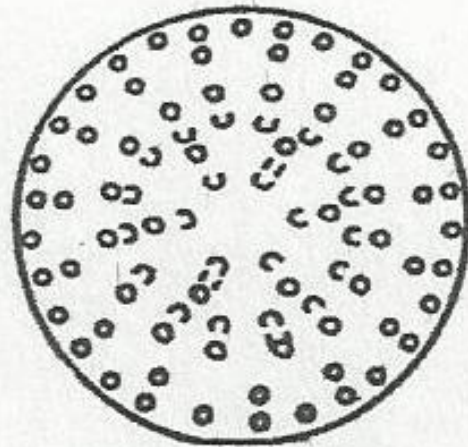
Wedge cut



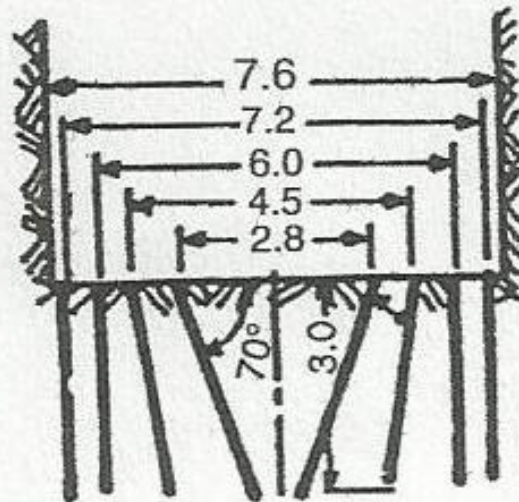
Rock: shale



Pyramid cut



Rock: sandstone



(b)

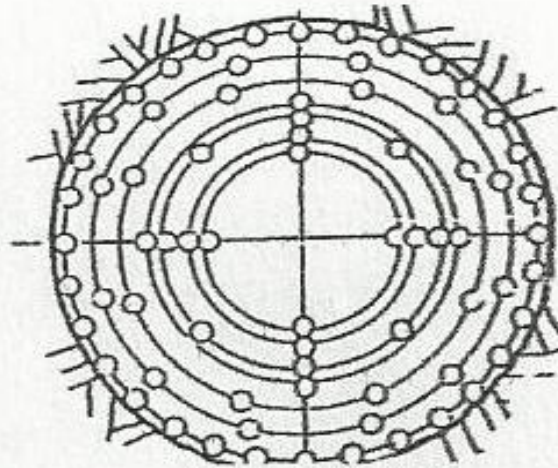
Drilling

- By pneumatic drill – rotary percussive
- Air pressure > 80 psi 
- Hollow drill steels fitted with detachable bits tipped with tungsten carbide
- Shape of drill bit depends on strength of rock
- No. of drills used / jumbo

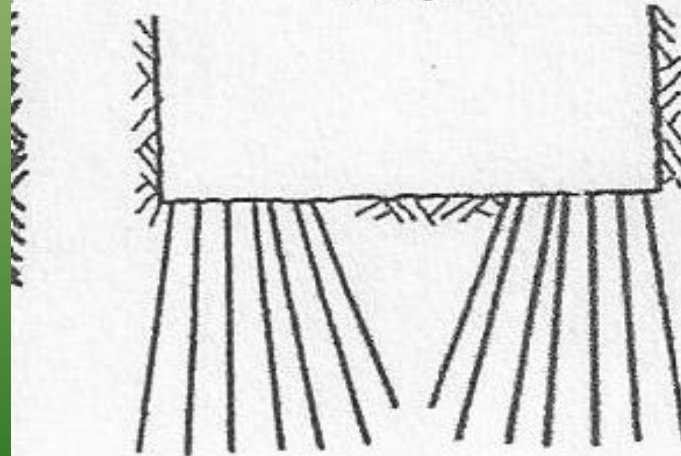
Arrangement of holes

- Concentric rings
- Cut holes at an angle towards the centre
- Sumpers / Inner easers / Outer Easers / Side holes or trimmers or flankers
- Angles of holes reduces towards the outer rings and side holes are vertical

Pyramid cut with jumbo

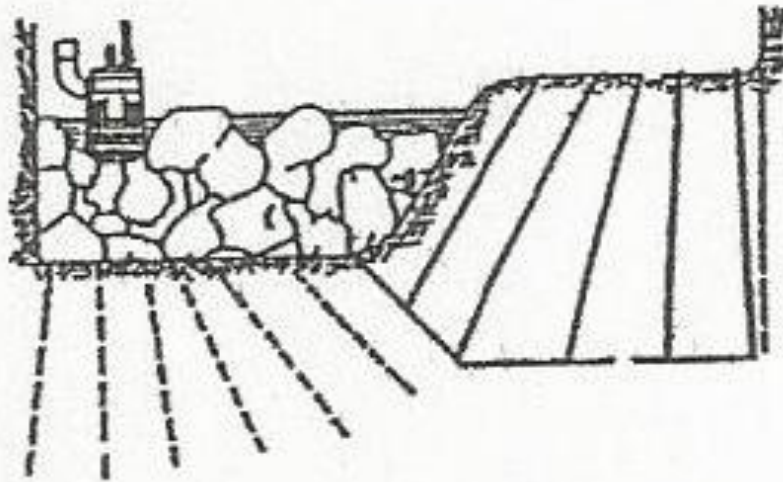
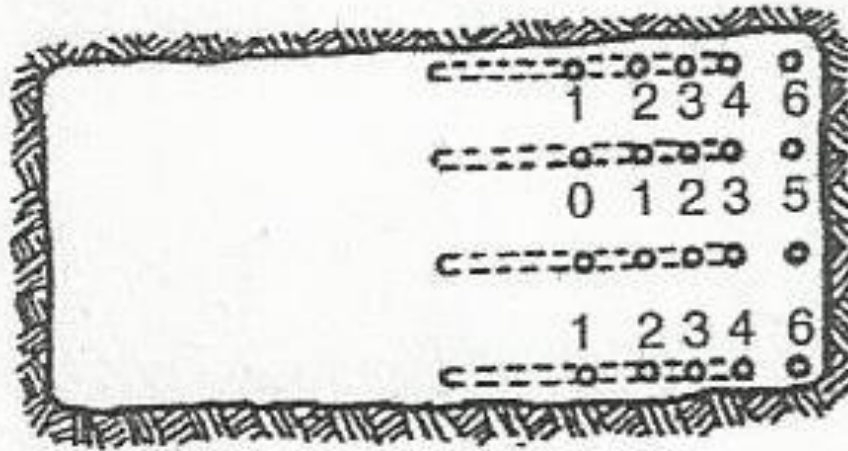


Pattern drilled
with jumo



(c)

Step cut



Step cut
(d)

Longer hole blasting

- Earlier holes were about 4 feet long
- Now holes of 4 to 5 meter length are drilled by jumbos
- Cut holes of 3.5 inches to 7.9 inches are drilled to provide initial free face
- ☐ • Emulsion explosives are less expensive, faster in loading, providing full borehole coupling, reduces drilling time and results better fragmentation

Blasting

- High density water resistant explosive such as Nitro-glycerin based explosive
- Aluminum based water gel explosive
- Water gel slurries
- Use of emulsion explosives with boosters and NONEL detonators

Blasting

- Weight of explosives per cubic meter of rock blasted depends on
 - Nature of rock
 - Strength of explosive
 - Degree of fragmentation required
 - Depth of pull
- No. of holes varies from 80 to 100
- Weight of explosives per hole
 - Total explosives
 - No. of holes
 - 750 gm – 1 kg per hole of 6 ft

Blasting

- Water or sand clay mixture as stemming
- ☐ • Stemming used – 3:1 sand clay mixture for dry shaft or sand alone in wet shaft
- Series – parallel connections
- Blasting cable from surface
- Main shot firing cable suspended from surface
- Must be able to support its own weight
- 2 core cable or 2 separate single core placed on opposite sides
- PVC insulated single wire armoured and sheathed
- Coiled on drum

Removal of debris

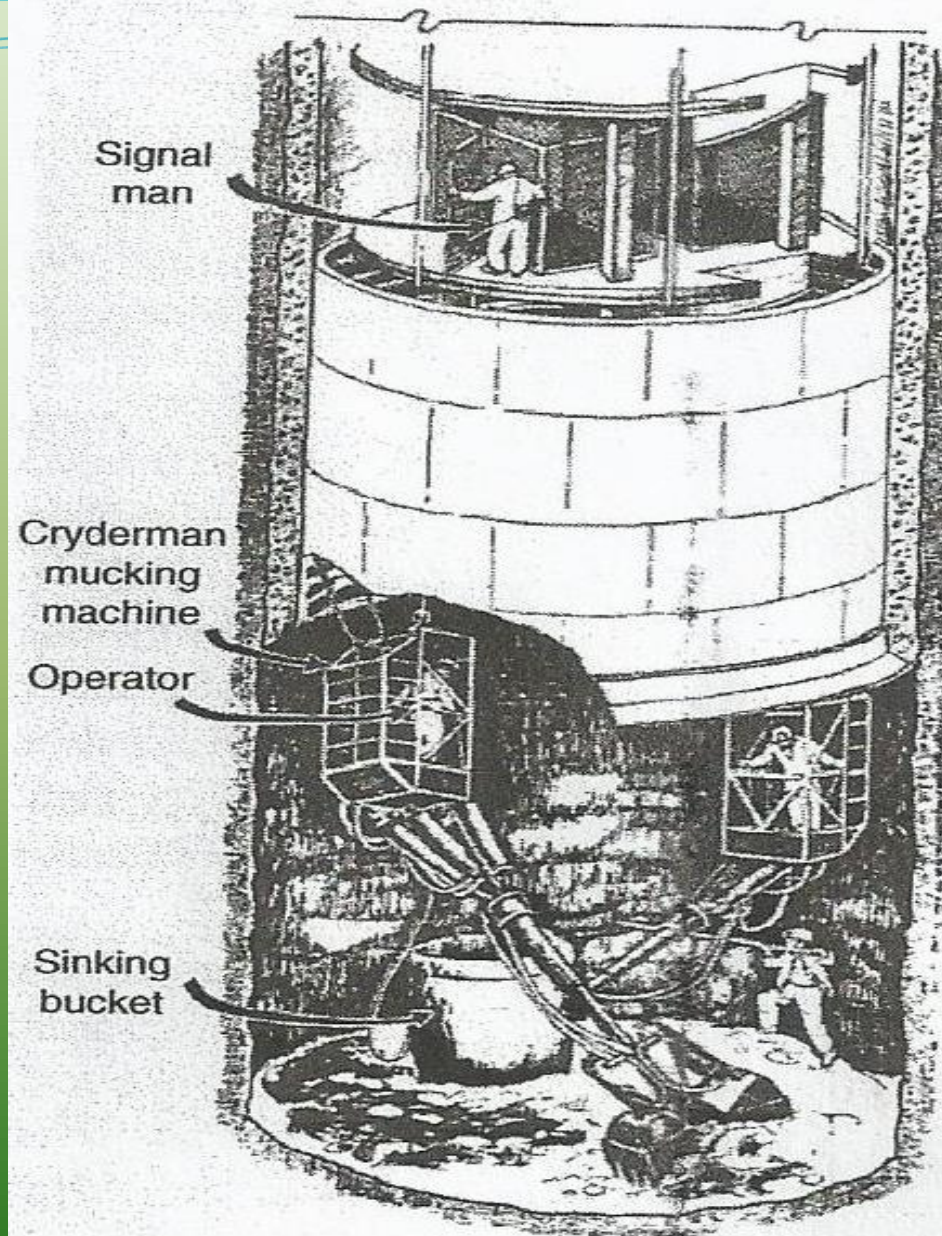
- Difficulties
 - Presence of water
 - Limited space
 - Time required for installing mucking equipment
 - Occupies about 50-60% of sinking cycle
- Mucking efficiency depends on
 - Size of rock fragments
 - Hoisting depth
 - Shaft cross section
 - Water inflow rate

Removal of debris

- Broken debris loaded into kibbles or bowks, manually or mechanically
- More than one kibbles are used at shaft bottom
- Mechanical loaders use compressed air operated grab
- The grab consists of a ring of powerful steel jaws or fingers which can be opened or closed by comp air operated by a man at shaft bottom
- The grab hangs from a control tower or frame mounted on the permanent lining
- The tower can be raised or lowered by a winch at the surface

Lashing and mucking

- Lashing unit is a mechanical device incorporating hoisting, slewing and radial traversing mechanism for handling of the cactus grab
- Types of shaft mucker
 - Arm loaders like cactus grab / cinderman mucker / backhoe mucker
 - Rocker shovel
 - Scraper

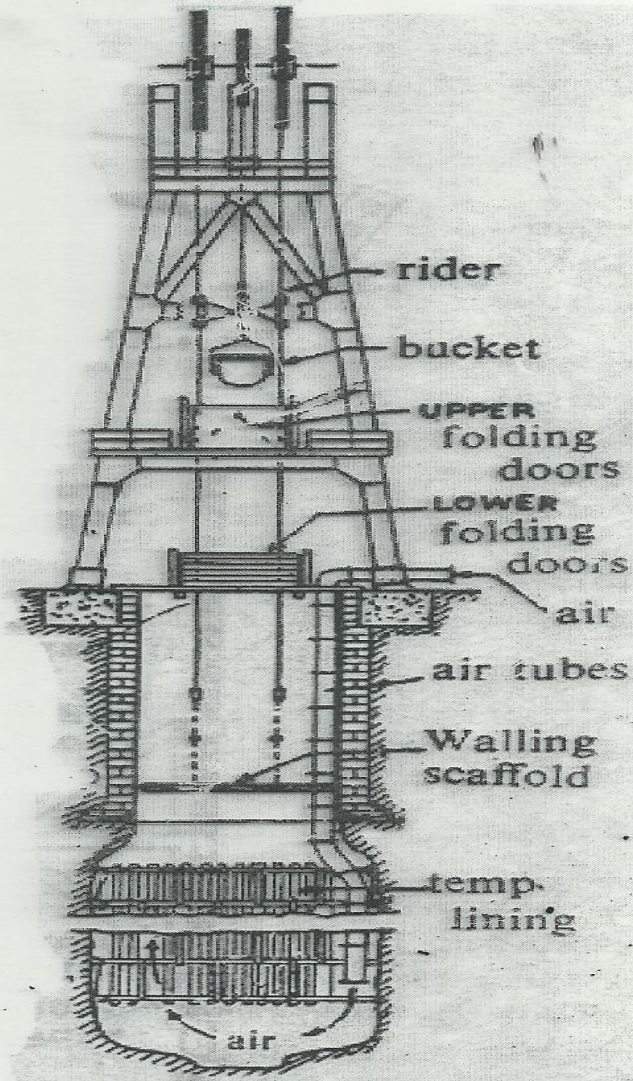


Cinderman
mucker

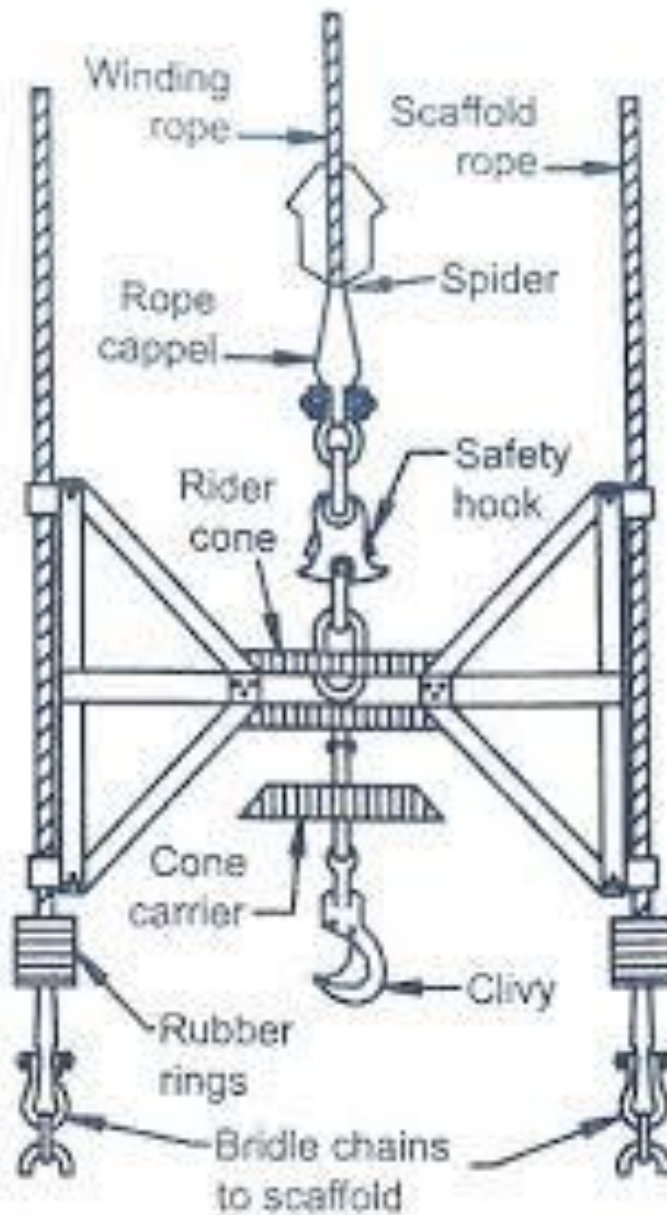
(a) Mucking using Cryderman mucker

Hoisting

- Hoisting / Lowering of men, material and muck by temporary hoist, headgear and other attachments
- Main components
 - Head gear with pulleys
 - Substantial construction
 - Capable of withstanding load upto 120-150 te
 - Compact to enable erection of permanent structure
 - Bolted construction for easy dismantling
 - Central pulley for sinking rope
 - Two side pulleys for walling platform ropes
 - All three ropes are locked coil, non-spinning type
 - Platform ropes also act as guide ropes for rider to keep the kibble steady in the shaft when being raised or lowered
 -



General Arrangement
of Sinking Shaft.

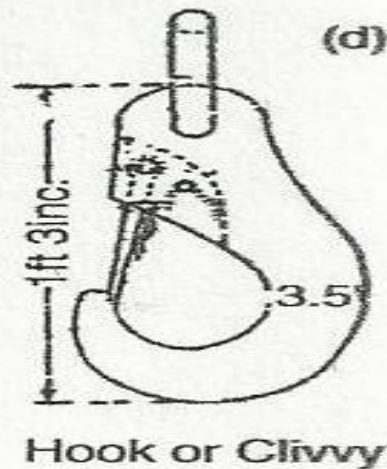
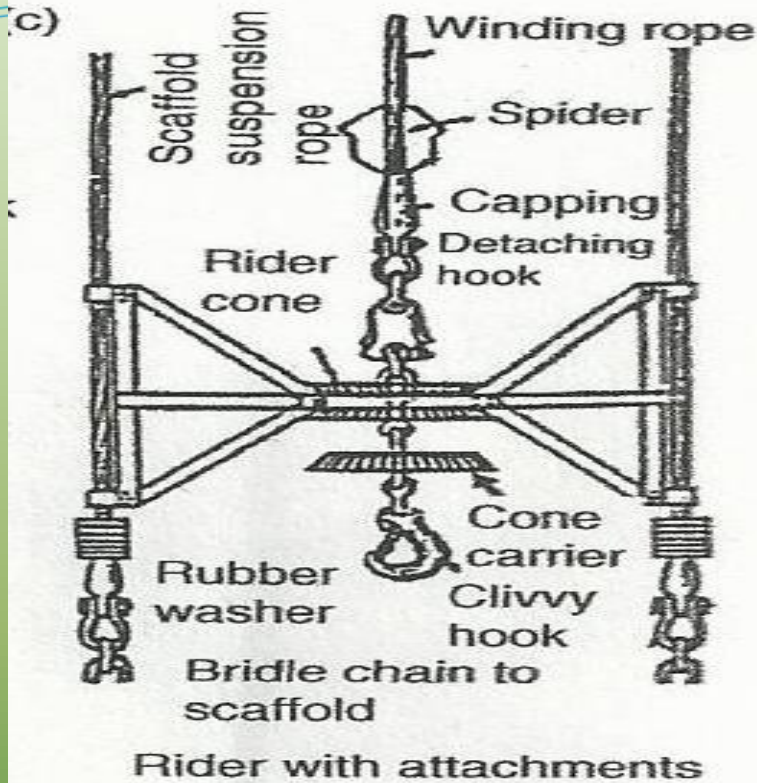


Hoisting

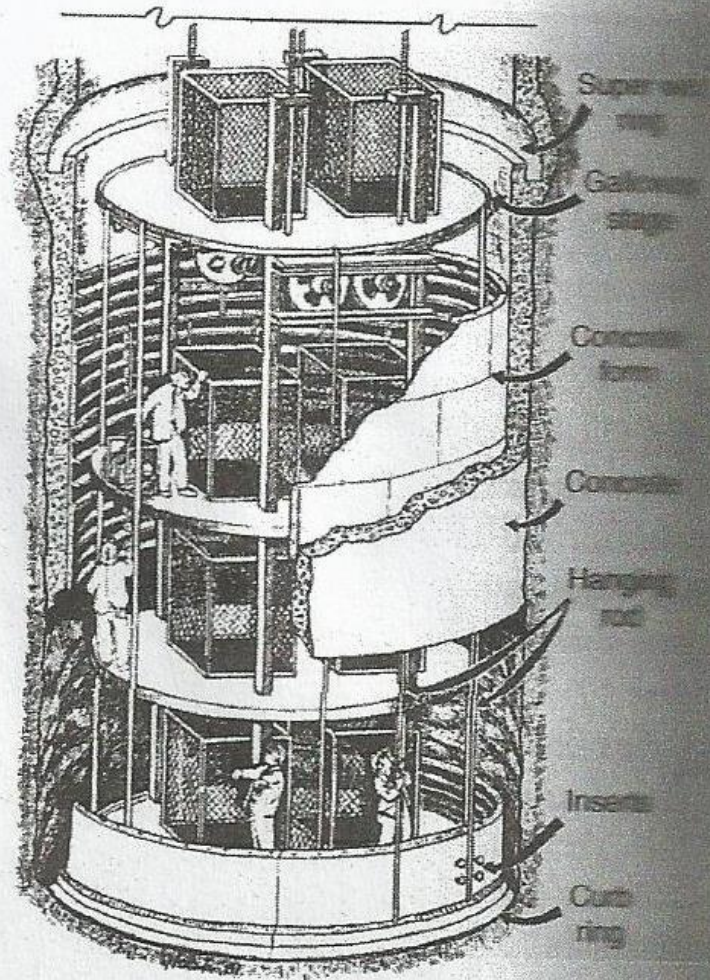
- Hoisting / Lowering of men, material and muck by temporary hoist, headgear and other attachments
- Main components
 - Rider
 -  • Permanent guide ropes are not possible to be installed while sinking in progress
 - When a walling is used, the scaffold ropes with an appliance RIDER may act as guide ropes for the kibble above the point where scaffold is suspended in the shaft
 - Davis and Barker's Cone sinking rider is well known
 - The vertical limbs should be long enough to prevent canting of the rider

Hoisting

- Hoisting / Lowering of men, material and muck by temporary hoist, headgear and other attachments
- Main components
 - Rider
 - At the centre of the cross bar of the rider, a circular block with hole large enough for the detaching hook to pass through to prevent over winding
 - Mounted loosely on the winding rope is a collapsible spider or guide sleeve which fits into the central hole (and remains there, so keeping the rope steady) when the rider is at rest on the walling scaffold and the kibble suspended from the spring hook, below that level
 - When the kibble is above the scaffold, rider is lifted by the carrier cone inserted in the bridle chain, which engages in the underside of the block
 -



GENERAL ARRANGEMENTS FOR SHAFT SINKING



(b) Supporting using concrete

upporting system using concrete.

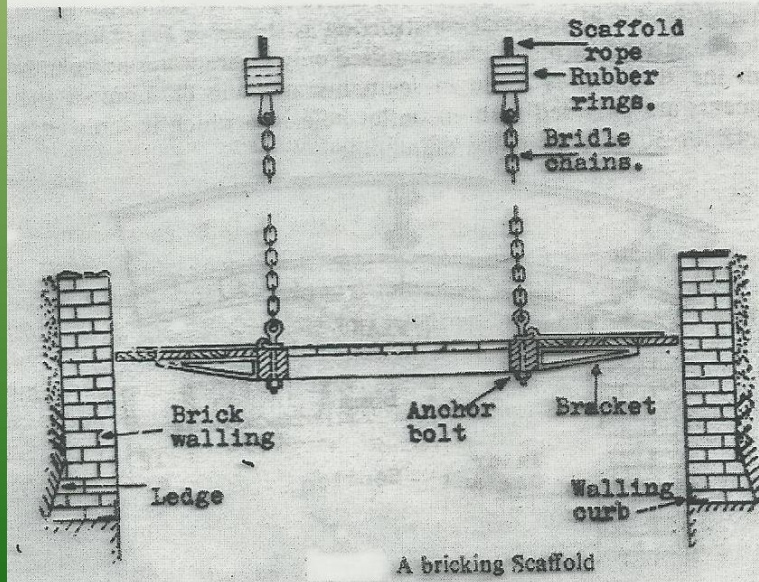
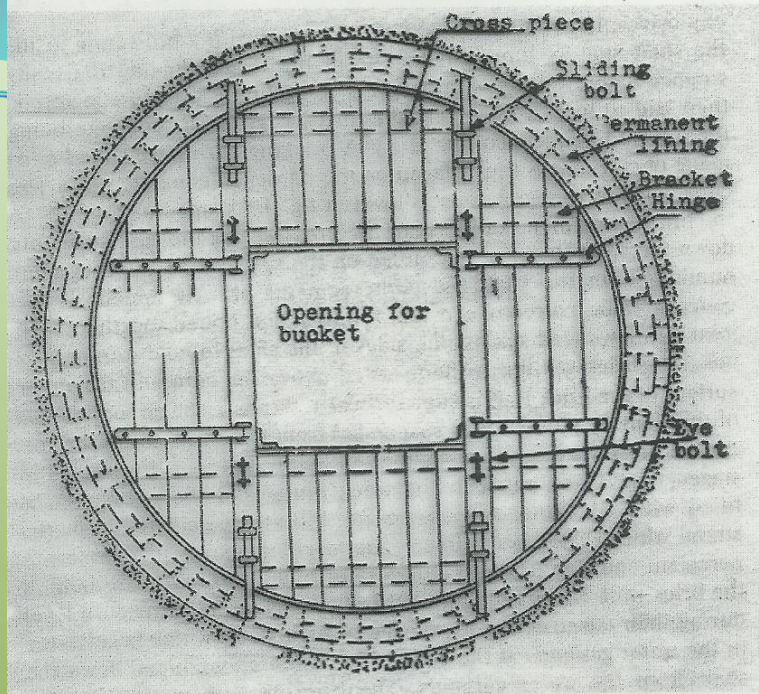
Walling platform

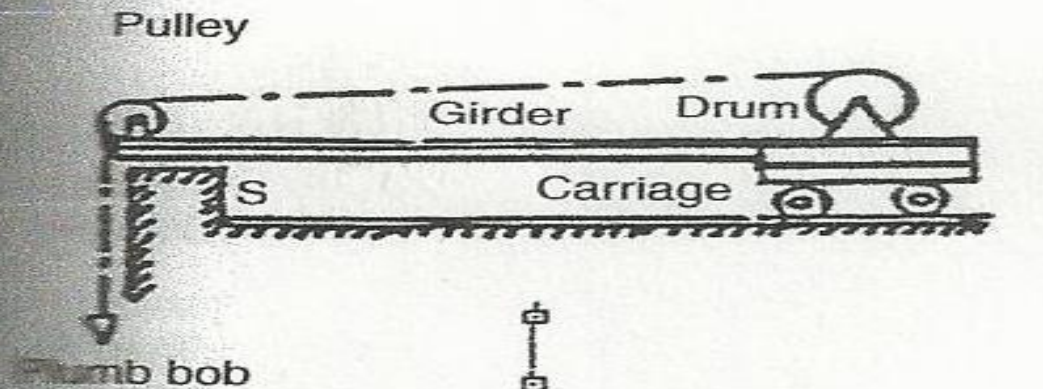
- Used for construction of side walls
- Dia of the platform is slightly less than dia of the shaft
 - Suspended by two ropes, which also act as guide ropes for the kibble
 - Each rope carries two bridle chains secured by shackles to four bolts
 - A no. of rubber rings, supported by clamps, are secured to each rope to relieve the impact of ride
 - The platform has a central opening for passage of the kibble
 - To keep the platform steady when persons are at work, four stout iron keys or bolts are pushed out to rest on the top of finished section of walling or in holes left in walling

Walling platform

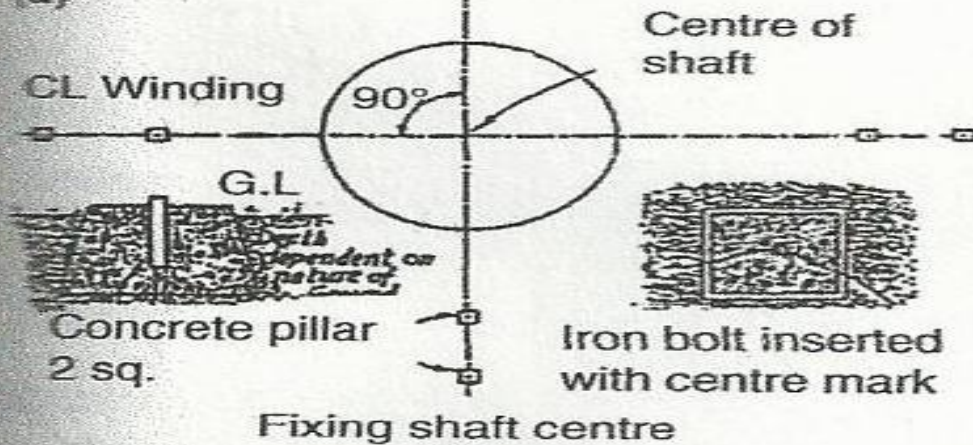
- During sinking, scaffold is suspended within 50 ft above shaft bottom
-  • The scaffold should be as close to the shaft bottom to guide the kibble
- When blasting, scaffold should be raised temporarily
-  • Besides being used for walling, scaffold also provides protection against falling materials
-  • It may also be used as emergency means of exit in case of anything wrong
- No of rope ladders are attached to the scaffold reaching to the bottom

Walling Platform

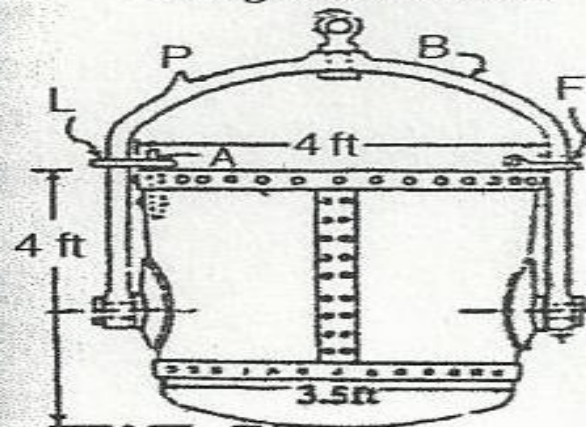




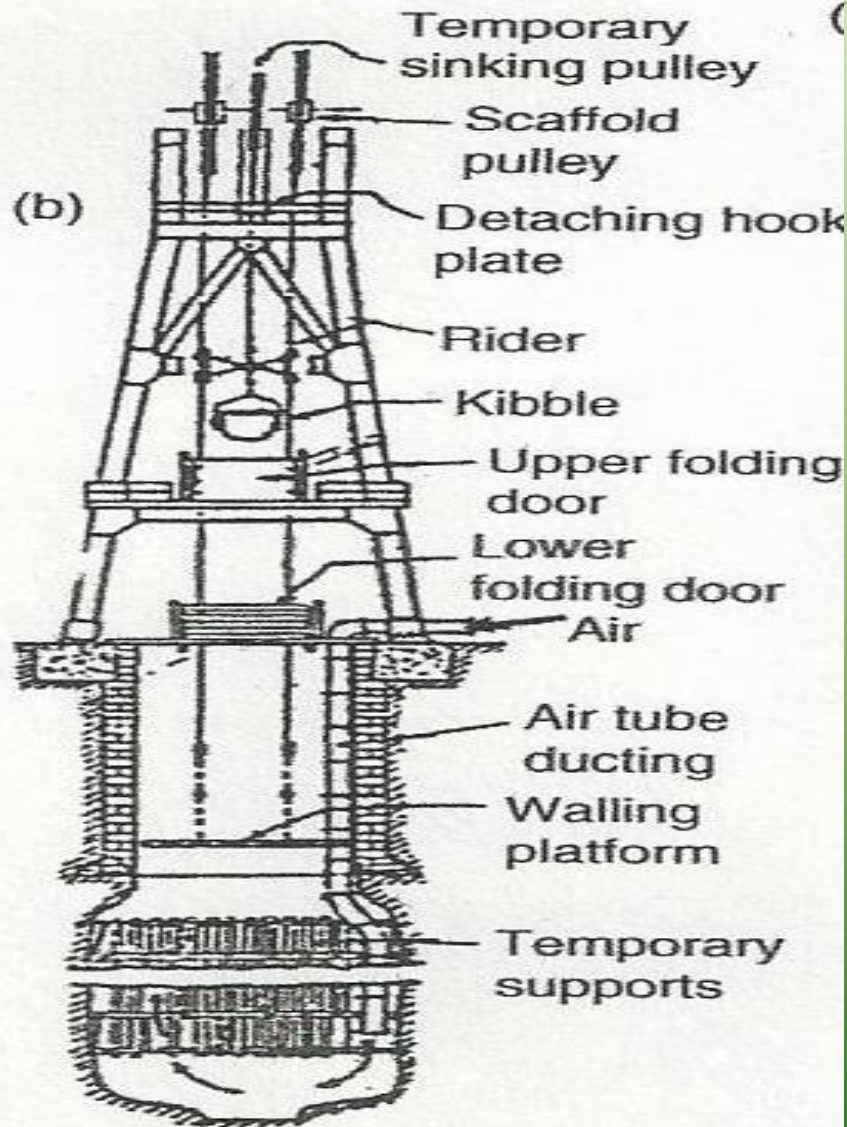
(a)



(e)

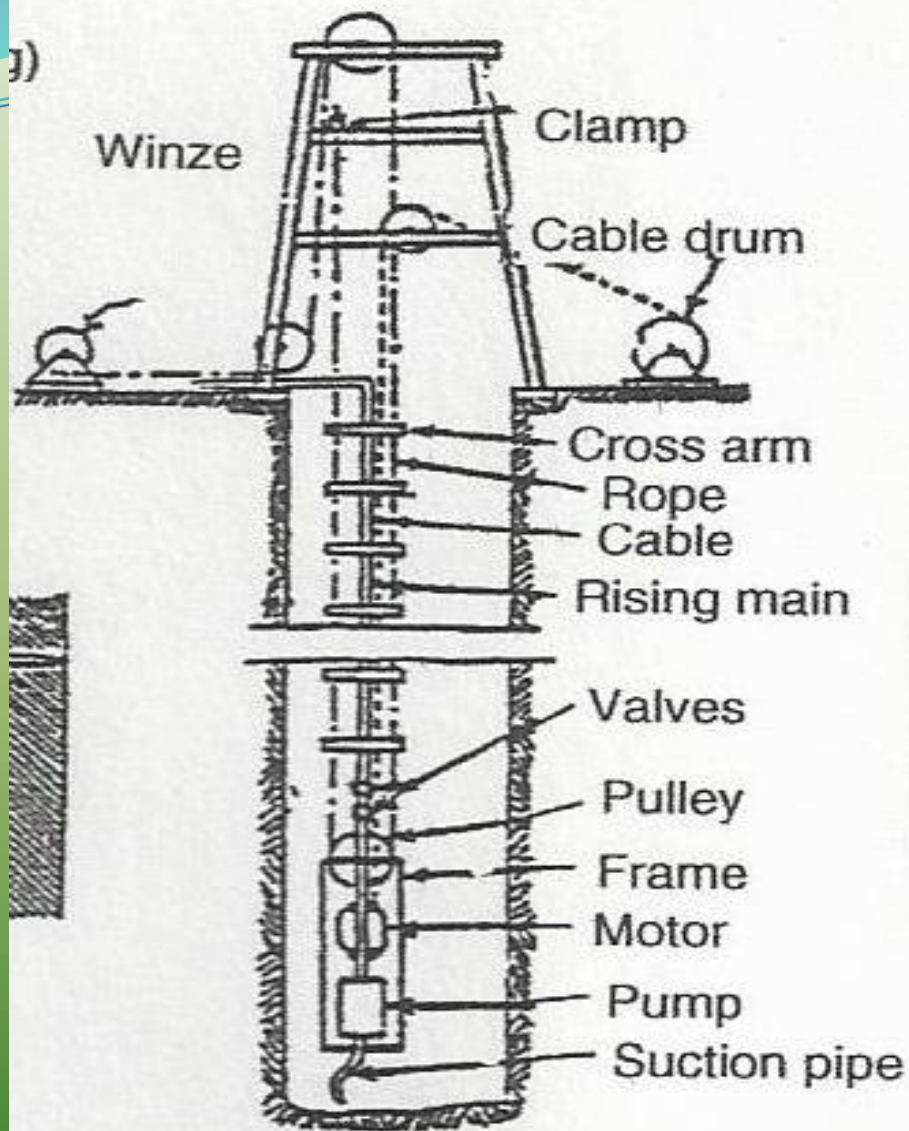


GENERAL ARRANGEMENTS FOR SHAFT SINKING



General arrangement during shaft sinking

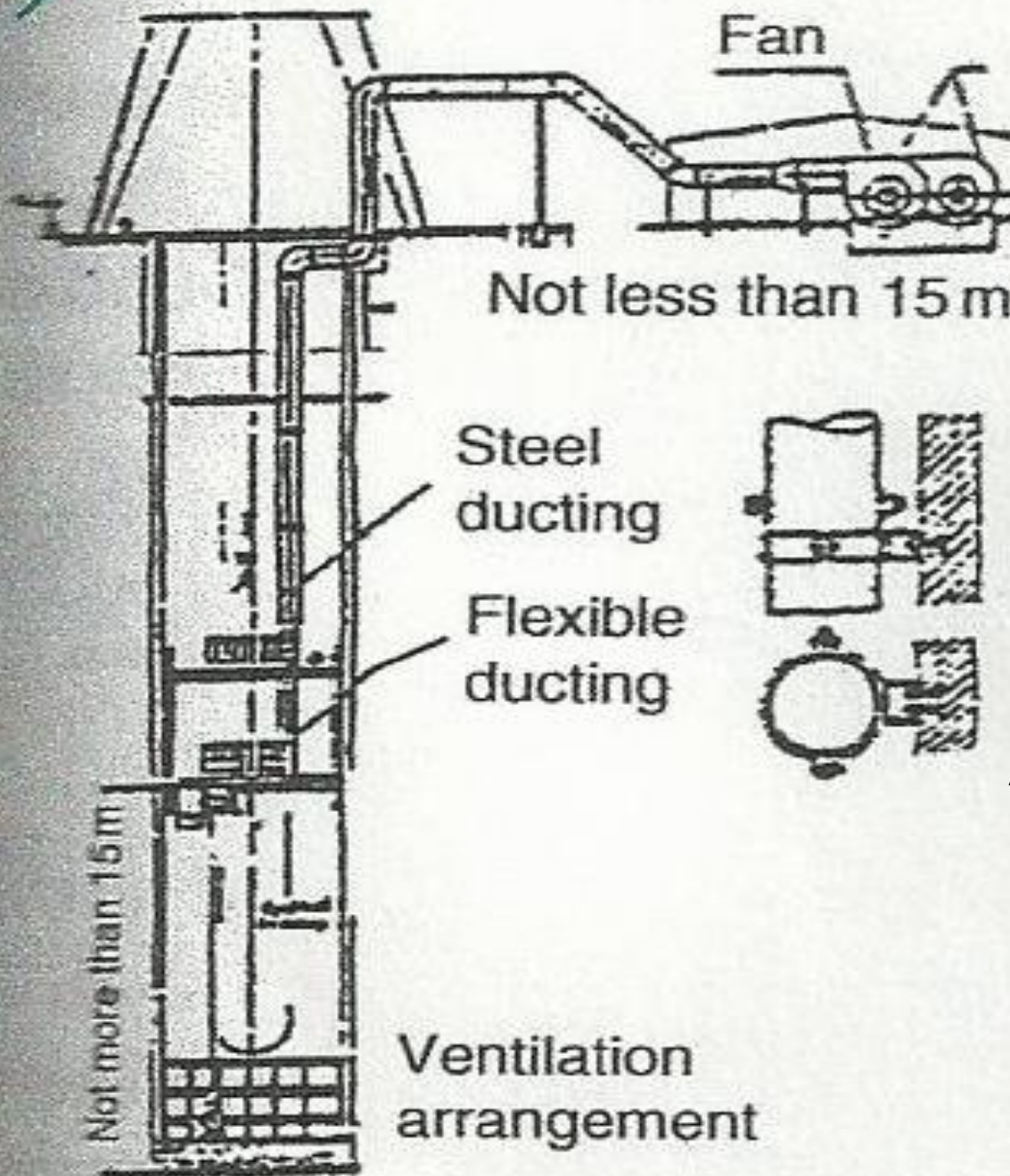
GENERAL ARRANGEMENTS FOR SHAFT SINKING



Pumpint arrangement
(Turbine pump's suspension shown)

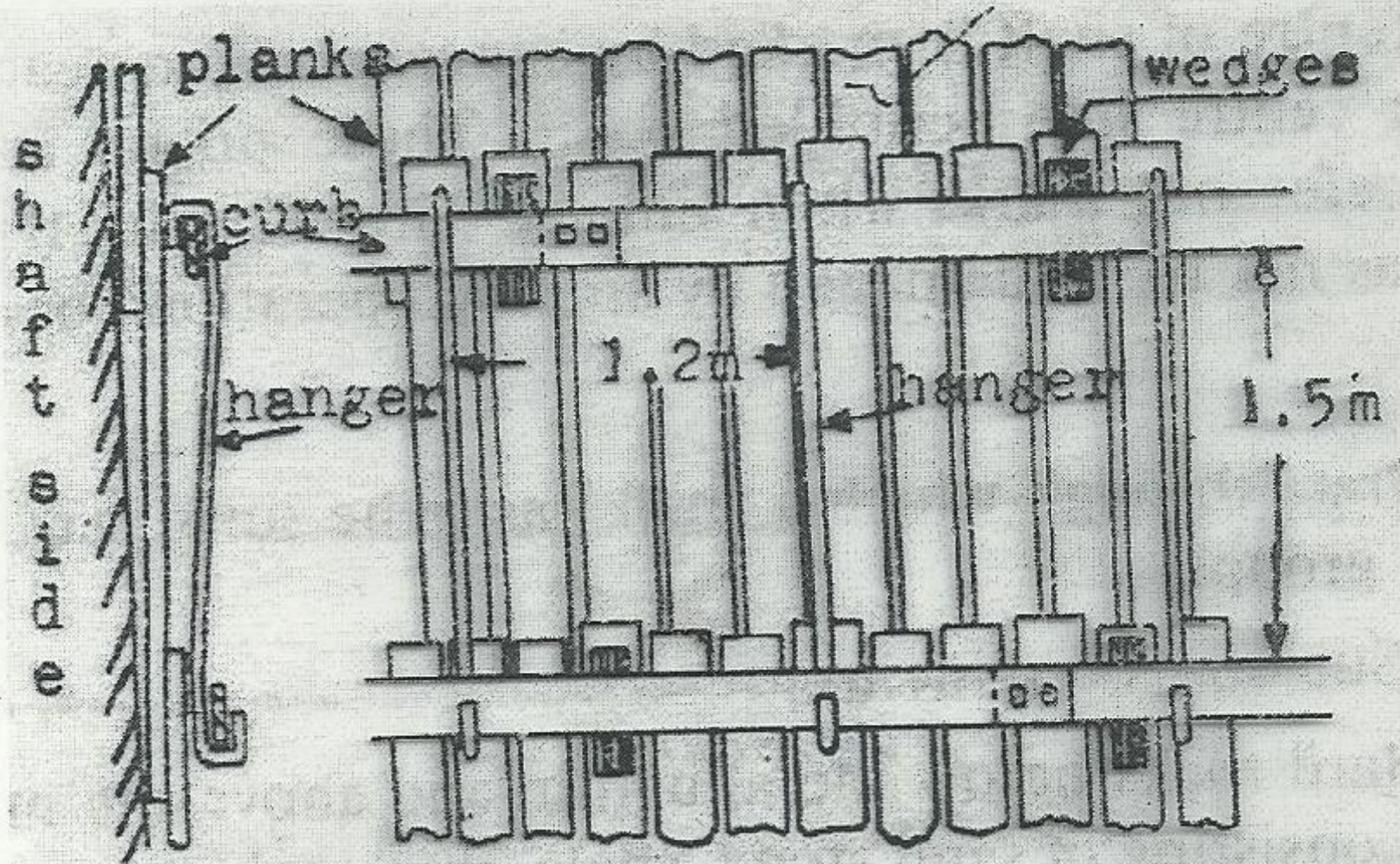
GENERAL ARRANGEMENTS FOR SHAFT SINKING

(F)



GENERAL
ARRANGEMENTS
FOR SHAFT
SINKING

Temporary support of sides



Temporary Lining

Temporary support of sides

- When one round of shots fired and debris removed, support of sides are necessary
- ☐ • Done by system of steel curbs or skeleton ring with close lining of wooden backing deals or steel plates
- ☐ • Mild steel curbs of 4-5 inch deep, 1 inch thick and 6-8 ft long segments with overlap and joined by bolts or fish plates
- Each curb is suspended from the one above by 6-8 s-shaped steel hooks or hangers
- Every third or fourth ring must be additionally supported on plugs placed into holes horizontally drilled into the strata
- Wooden backing deals about 5-6 ft long are placed close together behind the curbs and each deal is provided with its own wooden wedge to tighten the structure

Temporary support of sides

- In all steel arrangements, curbs are channel sections and are butt jointed by fish plates and four steel pegs
- Curbs are supported by hangers and are backed by steel plates provided with a hook enabling them to be suspended behind the curbs
- ☐• Very quick to install and can be reused many times
- ☐• Temporary supports are removed, section by section, as the permanent lining is built up.

Permanent lining of shafts

- Types of shaft lining
 - Brick walling – dry shafts
 - Concrete blocks -
 - Monolithic concrete – Most commonly used
 - cast iron tubbing – heavily watered shaft
- Function of shaft lining
 - To prevent deformation
 - To prevent decrepitating due to influence of atmospheres
- Thickness of lining
 - Stresses to be resisted
 - Nature and strength of material used for lining

Permanent lining of shafts

- Stresses on shaft lining
 - Compressive in nature – from outside to inside of shaft
 - Depends on
 - Dia of shaft
 - Depth of shaft
 - Nature of strata
 - Presence of water
 - Tensional stress may be developed locally due to presence of faults, cavities, or other weaknesses

Permanent lining of shafts – Brick walling

- Suitable for compact and fairly dry strata, where stresses are not high
- 14 inch walling is adequate
- Often 9 inch walling is sufficient
- Lime mortar or cement mortar is used

Walling curb

- Each length of brick work is built up from a walling curb laid on a specially prepared bed of strong ground
- The curb may be of cast iron or concrete
- Cast iron curb is a L shaped ring, 10-18 inch wide and $\frac{3}{4}$ to $1\frac{1}{2}$ inch thick, divided into number of segments
- Ends of segments have vertical flanges, for bolting adjacent segments
- Curb bed must be specially prepared, smooth and level
- Curb should be truly circular and horizontal
- Its centre should coincide with the shaft centre

Cast iron walling curb

accrete.

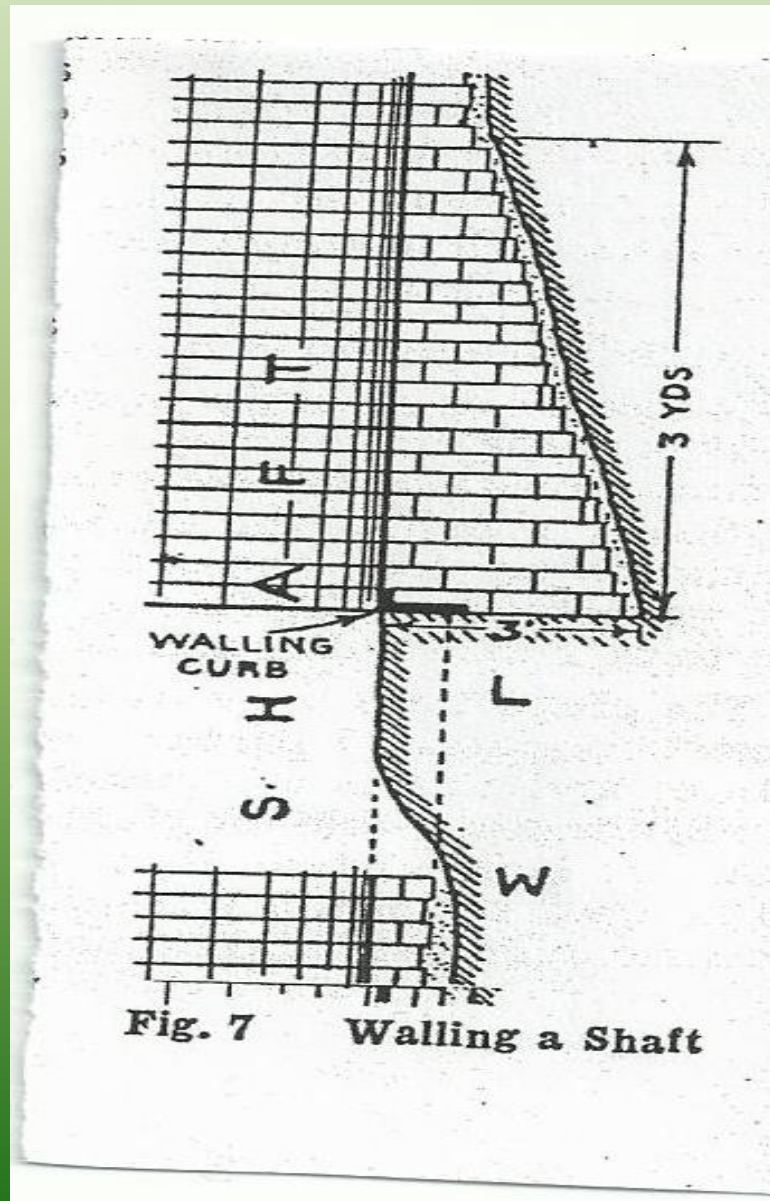


Fig. 6 Cast Iron Walling Curb

Erection of brick walling

- When approaching the site of the curb, size of shaft is gradually increased or laid back about 2 ft all round
- ☐ • Inside dia of the curb is the finished dia of the shaft
- Walling is built up from the curb, beginning about 2 ½ to 3 ft thick and gradually decreasing to normal thickness
- To maintain verticality, side plumb lines are suspended from the curb above
- Temporary supports are removed in sections as walling ascends
- ☐ • All spaces behind walling are packed solid with ashes, broken rocks or bricks or with concrete

Walling a shaft



Erection of brick walling

- When a length of walling is completed, sinking is continued at a reduced diameter, in alignment with the inside of walling
- Shaft is then widened out to its ordinary size, some 2-3 m below, to leave a strong ledge of stone, as at L, to support the walling
- The thicker portion serve to divide the entire shaft wall into separate portions independent of each other, each carrying out its own weight
- If ever one portion is collapsed or had to be removed, other parts would not be disturbed

Erection of brick walling

- When next lower length of walling is built up, it is continued of the required thickness until it reaches the point W
- The front portion of the ledge beneath the last curb is then cut away, a narrow width at a time, but upto the full height, so as to make room for the remainder of the walling
- ❏ • This is called underpinning of the walling curb

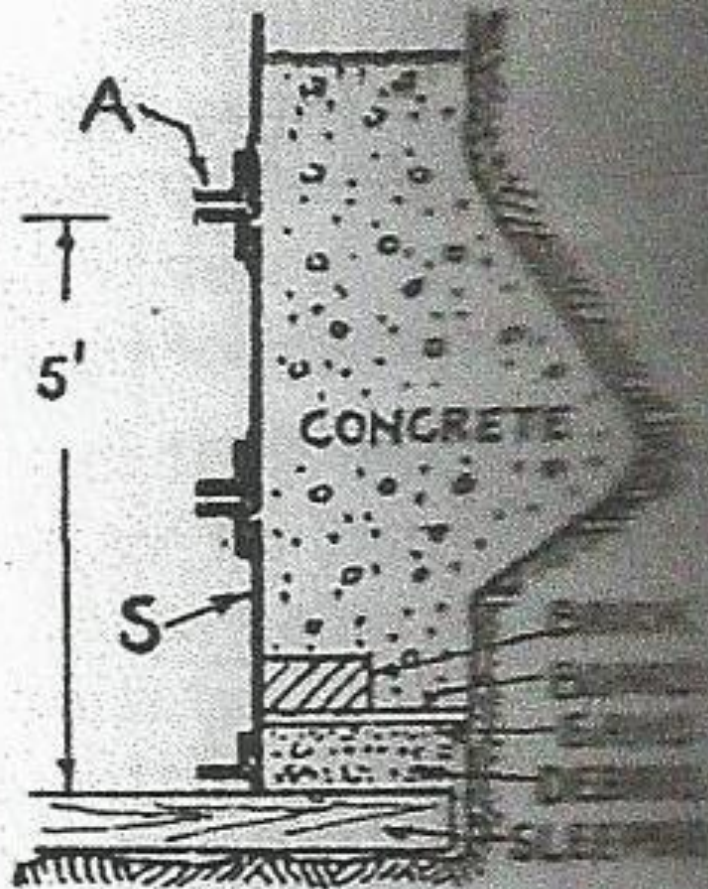


Fig. 8

Concrete Walling

Concrete walling curb

- In this case, the ground is cut back about 2 ft in wedge form so as to key the concrete curb into the strata
- To form the curb, to retain the wet or plastic concrete in position, shuttering is placed
- Shuttering (S) are steel sheets, curved to suit the circumference of the shaft and having angle irons A riveted to them to enable adjacent segments to be securely bolted together

Concrete walling curb

- At the shaft bottom, each segment of the first shuttering ring rests on a sleeper
- The complete ring, when bolted up, must be carefully leveled and centered
- Small debris of 6 inch height is placed behind the ring, followed by a layer of sand covered with brattice cloth
- One or more rings of bricks may be laid so that they may be removed later to enable the length of the walling to be keyed into the above

Concrete walling curb

- Concrete is now placed behind the shuttering ring
- Is well rammed to fill all cavities next to the strata and prevent formation of air pockets in finished concrete
- The second ring of shuttering is then placed, centred, leveled and filled up same way
- Temporary lining is removed as concreting procedes

Concrete walling curb

• Advantages of concrete curb

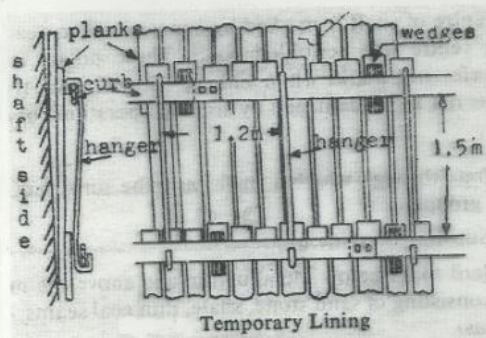
- They are self supporting and underpinning is not required
- They can be applied in any ground
- Less liable to damage from blasting when sinking is in progress
- Very strong
- Long length of walling can be carried on a single curb

Monolithic concrete lining

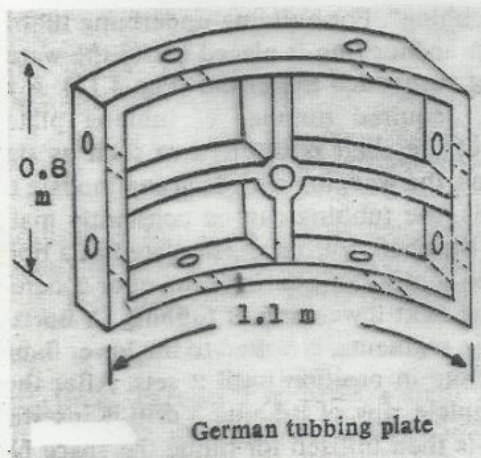
- ☐ • Monolith is a pillar or column consisting of single stone
- Monolithic concrete lining is one built up with a single mass of concrete instead of with concrete blocks

Monolithic concrete lining

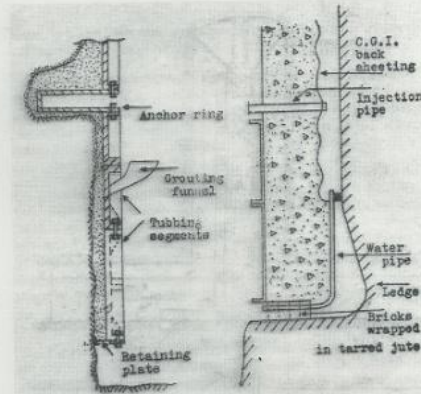
- Advantages
 - Can be constructed rapidly and at low cost
 - Higher Compressive strength (3000-5000 psi) compared to brick lining (800-1500 psi)
 - The concrete extends right back to solid strata, filling up all cavities and irregularities and this adds to the rigidity and strength of the lining
 - It presents a smooth surface to ventilating current
 - Suitable for water bearing strata owing to its capacity to settle under water , its freedom from joints and its ability to withstand pressure when injecting liquid cement behind to seal off feeders
 - It can be rendered immensely stronger by steel reinforcement, where necessary



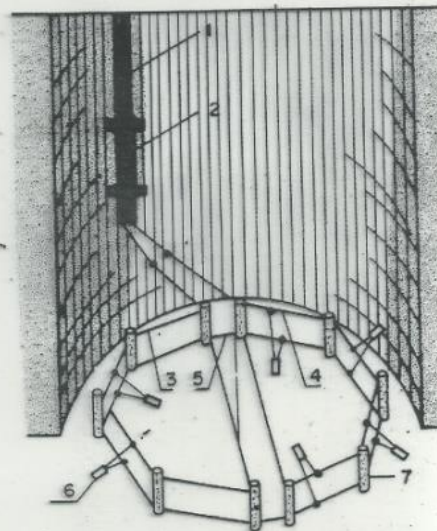
Temporary Lining



German tubbing plate

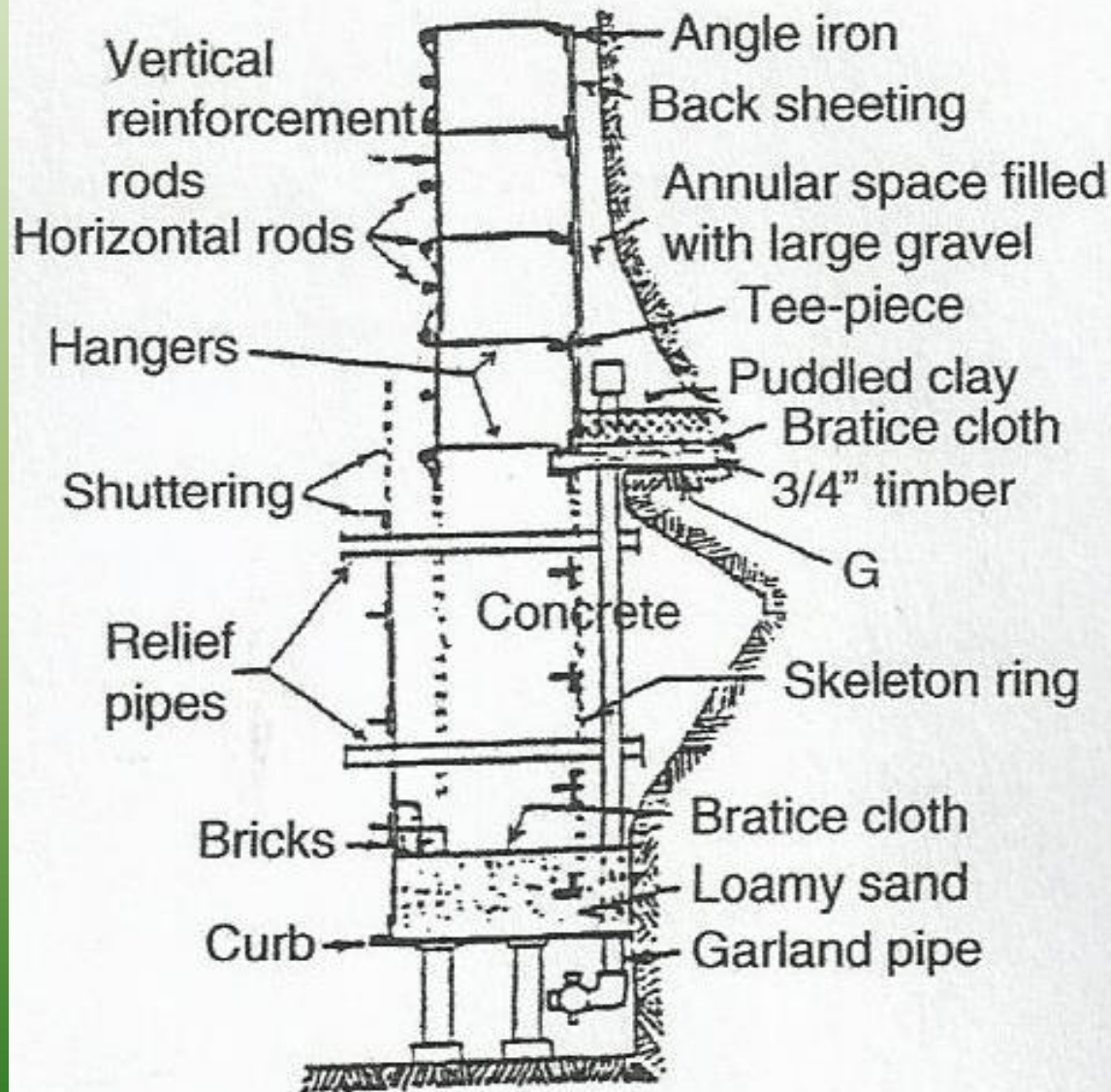


Left—Underhung tubbing
Right—Concrete lining of a sinking shaft.



Three-ring connection of electric caps.

1. shot cable
2. protecting pipe
- 3-4. active antenna
5. passive antenna
6. electric caps (in holes)
7. wooden stick supporting antennas.

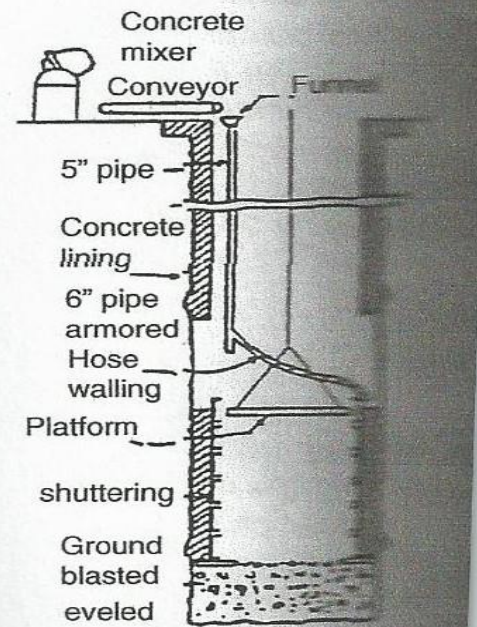


Permanent lining

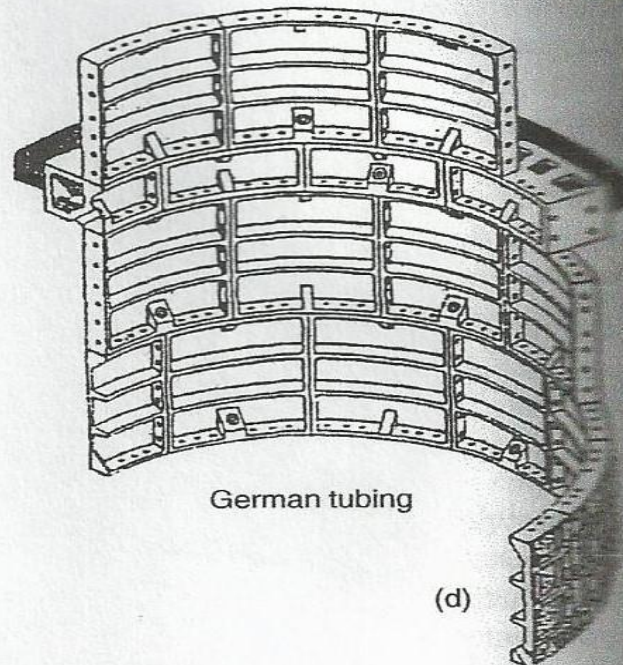
(a) Reinforced monolithic lining

t sinking

d



(b) Concrete walling



(d)

Erection of concrete walling

- A round of shots, below the length to be lined, is fired and the broken rock roughly leveled and left in position in readiness for resumption of sinking
- First shuttering ring is then truly centered, leveled on sleepers or wooden blocks
- A base is prepared for the concrete behind the ring
- When the length of shaft to be lined has been sunk and temporarily supported, a concrete curb is built up and shuttering rings are added progressively to retain the plastic cement

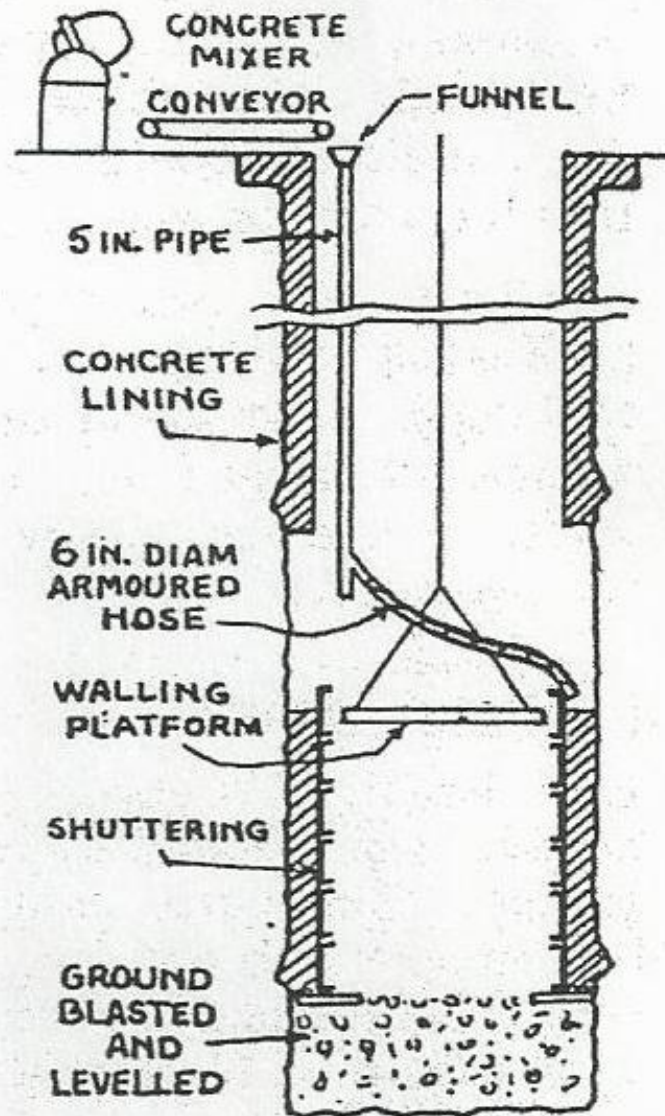


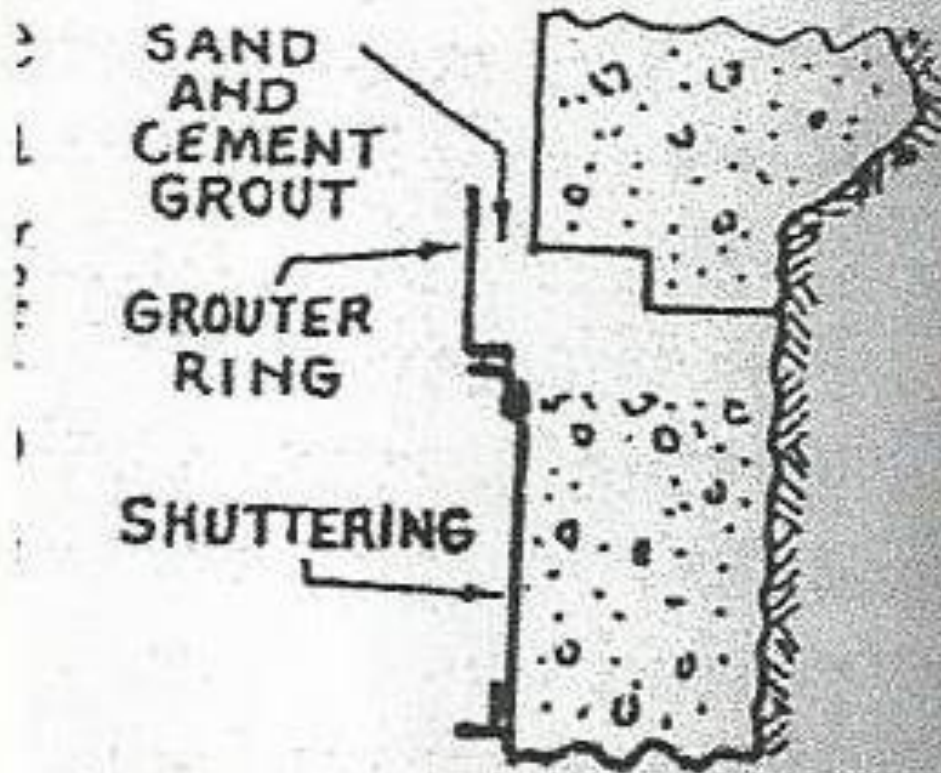
Fig. 9 Concrete Walling

Erection of concrete walling

- A second ring is added
- Concrete is fed from surface via pipe and armoured hose until the space behind the two rings is nearly full
- During filling, concrete is well rammed
- All temporary supports are removed in succession as work proceeds and further rings of shuttering added as required
- Shuttering must be left in position until such time concrete is set
- To facilitate the removal. Back of the shuttering are cleaned and well greased each time

Erection of concrete walling

- When the rising wall reaches the next section of wall above, care is needed to make a good joint
- A spl L shaped grouter ring is mounted on the last shuttering ring
- When concrete has reached within an inch or two of the top wall, fine grout of sand and cement is poured in to close the gap
- Grouter ring is filled to the top
- When grouter ring is removed, the excess material is dressed off



**Fig. 10 Connecting
Two Lengths of Walling**

Erection of concrete walling in water bearing ground

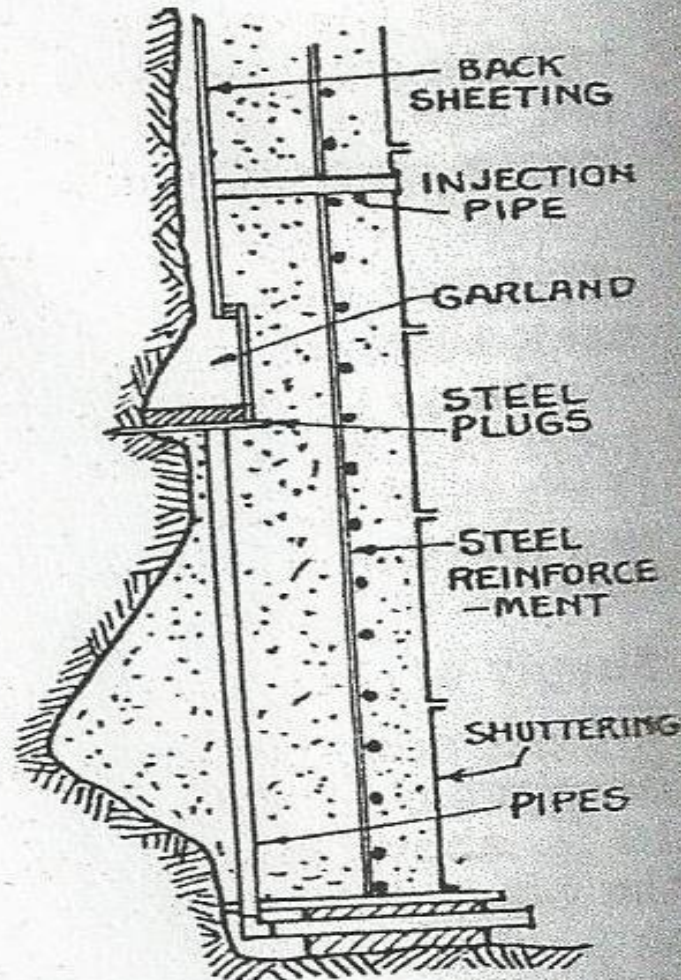


Fig. 11

Concrete Lining in
Water-bearing Strata

Erection of concrete walling in water bearing ground

- Back corrugated sheets are placed all round the shaft and are held in position by skeleton curbs and wedges
- These sheets serve the purpose of temporary lining and also of keeping water away from concrete during walling
- ☐ • At suitable point, where walling is to begin, a temporary water garland is formed to catch water flowing behind back sheeting and is led into pipes from which it is discharged down into the shaft

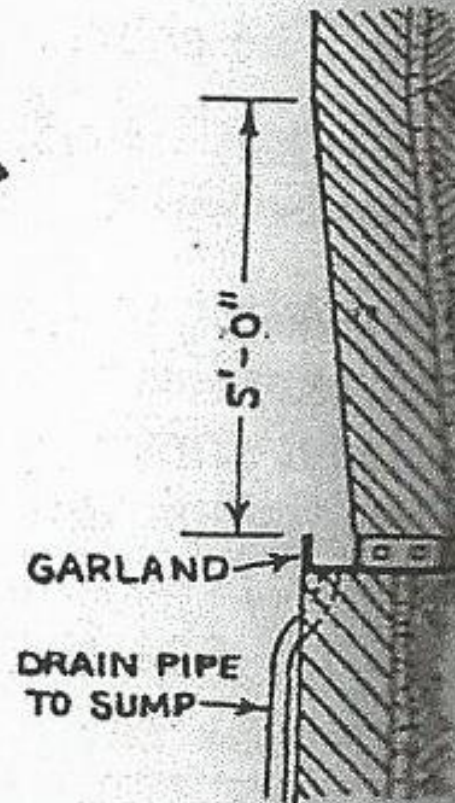
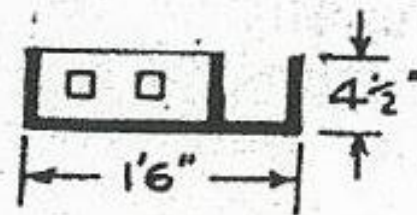
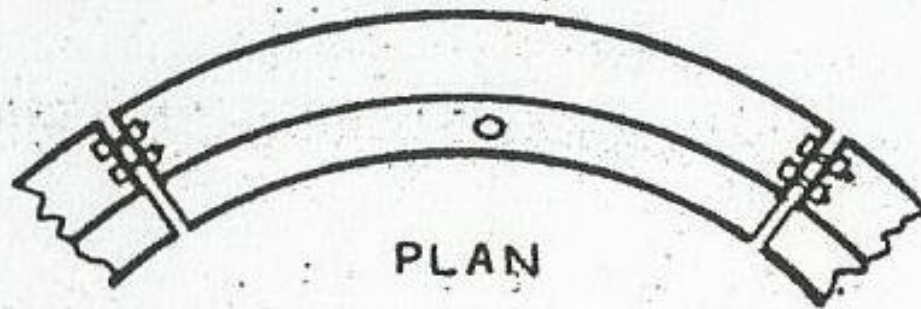
Erection of concrete walling in water bearing ground

- A base is prepared to receive concrete and first ring of shuttering and first length of steel reinforcement are erected
- Concreting is now begun, skeleton curbs being removed stage by stage
- Further rings of shuttering and steel reinforcement are added
- Space behind the back sheeting is filled with clean gravel by removing portions of the back sheeting at intervals
- Horizontal relief pipes are laid in the concrete at regular interval, extending through holes in the shuttering
- Later on, after removing the shuttering, fine grout of cement and sand is injected to seal behind the back sheeting

Cast Iron Tubbing

- An alternative method of permanent lining of shaft sunk through heavily watered strata
- Mostly associated with freezing process, where there is difficulty in setting of concrete
- Tubbing is inserted while the ground is still frozen, so that the sinkers work under dry condition
- Later, the ground is allowed to thaw slowly so that water pressure may build up gradually behind the tubbing

Water Ring Garland



Water-ring or Garland

pipes convey the water away to the sump.

Water Ring Garland

- Cast iron garland is having a channel around its inner circumference to catch the water draining down the shaft sides
- Garland consists of a no. of segments, which when bolted together, form a circle having same dia as finished dia of the shaft
- Each segment is an open-topped casting of 18 inch wide, 4 ½ inch deep
- One or more of the segments provided with an outlet hole into which is screwed a nipple from which 2 inch dia pipes convey water away to pumps at shaft bottom
- To avoid front edge of the space water-ring projecting into shaft space, the walling above the ring is laid back about 4 inch and is brought back gradually to normal dia above 5 ft above

Special Methods of Shaft Sinking

- When the strata through which shafts to be sunk include deposits of loose or unstable ground, such as mud, sand, gravel, or alluvium, or when they contain excessive quantities of water which can not be dealt with by sinking pumps, or when both difficulties occur together, it is necessary to adopt special methods of shaft sinking

Special Methods of Shaft Sinking

- Pile sinking or Piling
- Caisson sinking
- Freezing
- Cementation

Piling System

- Suitable for sinking through loose ground near surface
- Piles may be of either wood or steel
- The system consists of driving down into loose ground a circular lining of wooden backing deals, which are called piles
- Piles vary in length from 6 ft to 15 ft, are 3 inch thick and 6-8 inch wide
- Each pile is shod at bottom with an iron point or shoe
- Piles are driven down by heavy mallets , and are placed skin to skin to form a complete circular ring
- They are held in place by wooden curbs or rings, placed at intervals of 2 to 3 ft

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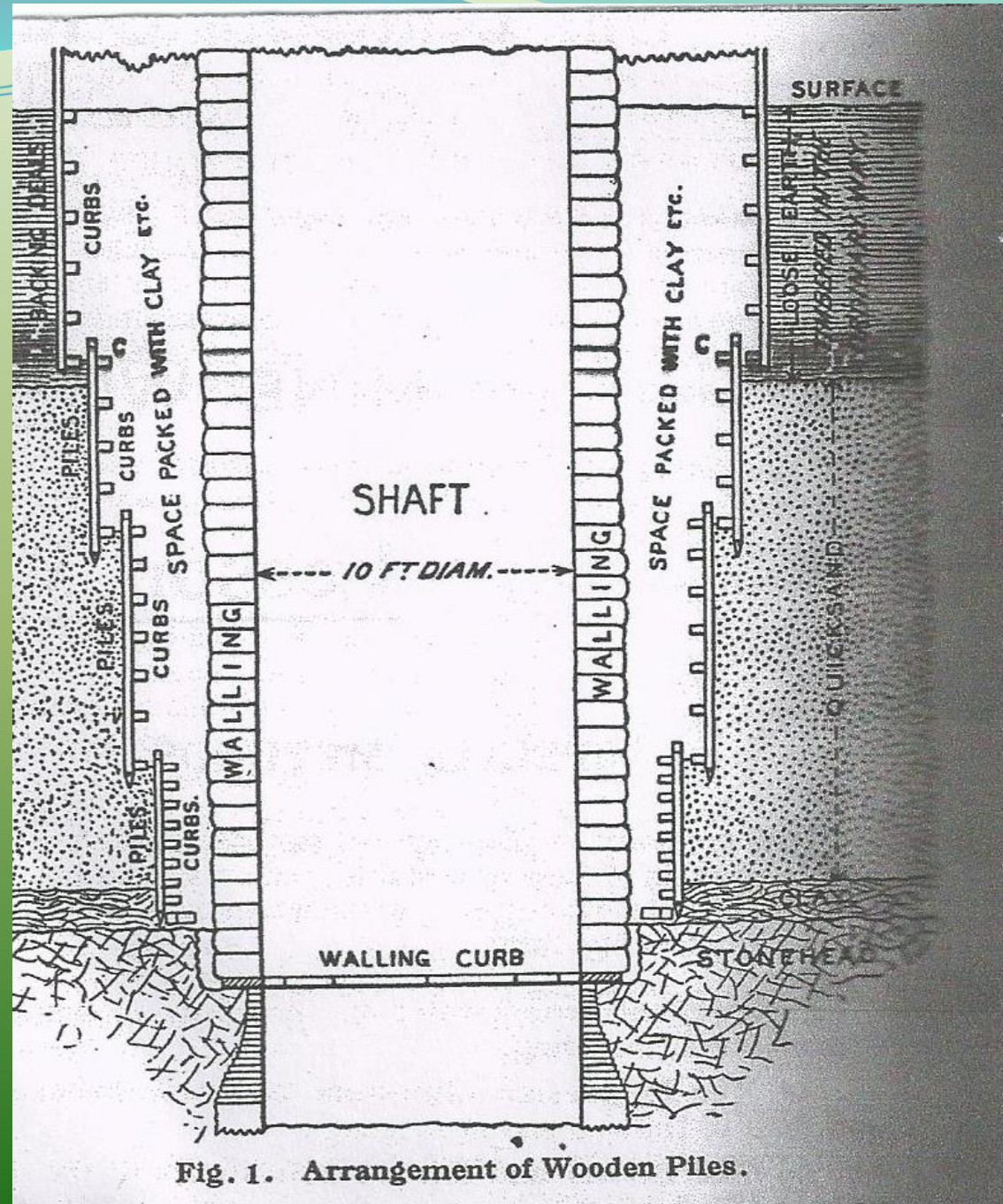


Fig. 1. Arrangement of Wooden Piles.

Piling System

- When the piles are driven a short way down, a curb is fitted within them
- The ground inside is cut away
- The piles are kept about 2 ft in advance of the excavation
- Piles are driven down one by one until hard ground is reached
- A walling curb is then laid and a strong wall is built up in front of the piles, the space behind suitably packed

Piling System

- Each successive ring of timbering is of smaller dia than the previous one
 - It is necessary to start with larger dia, which depends on depth of the running ground and dimension of the timber
-
- ☐ • For a shaft dia of 15 ft, with 8 ft long, 3 inch thick and 6 inch square curbs, starting dia is ?
 - There is a limit of depth beyond which it would be impracticable to use wooden piling system, as enormous excavation is required

Steel Piling System

- Much stronger than wooden piles
- Can be driven with much greater force with piling hammer

Caisson Methods

- Suitable for running ground at somewhat greater depths
- Three types
 - Sinking drum process
 - Forced drop shaft method
 - Pneumatic caisson method

Sinking drum process

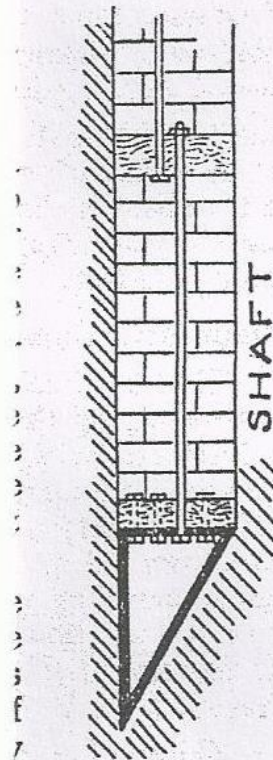


Fig. 3.
Section of
Brickwork
Drum.

Sinking drum process

- Similar to piling
- Lining of the shaft is formed in advance of excavation
- Piles are replaced by cylindrical drum of brick work and steel or of RCC, fitted with a steel cutting shoe at its lower end
- The brick work is 12-18 inch thick, resting on wooden curb fitted with steel cutting shoe
- Other wooden curbs are built into the cylinder at intervals of 3-4 ft, and are tied together by wrought iron rods to increase strength of the structure

Sinking drum process

- The drum sinks gradually by its own weight as the shaft is excavated
- Further brickwork is added at the top
- Care must be taken to ensure drum descends vertically
- Sometimes wooden boards are placed outside the drum to reduce friction

Sinking drum process

- Advantages
 - Lining is built on surface, where construction is safest and cheapest
 - Drum acts as permanent lining for finished shaft
 - Cost of temporary timbering is eliminated
 - Weight and strength of drum sufficient to push aside boulders etc that would stop timber or steel pile

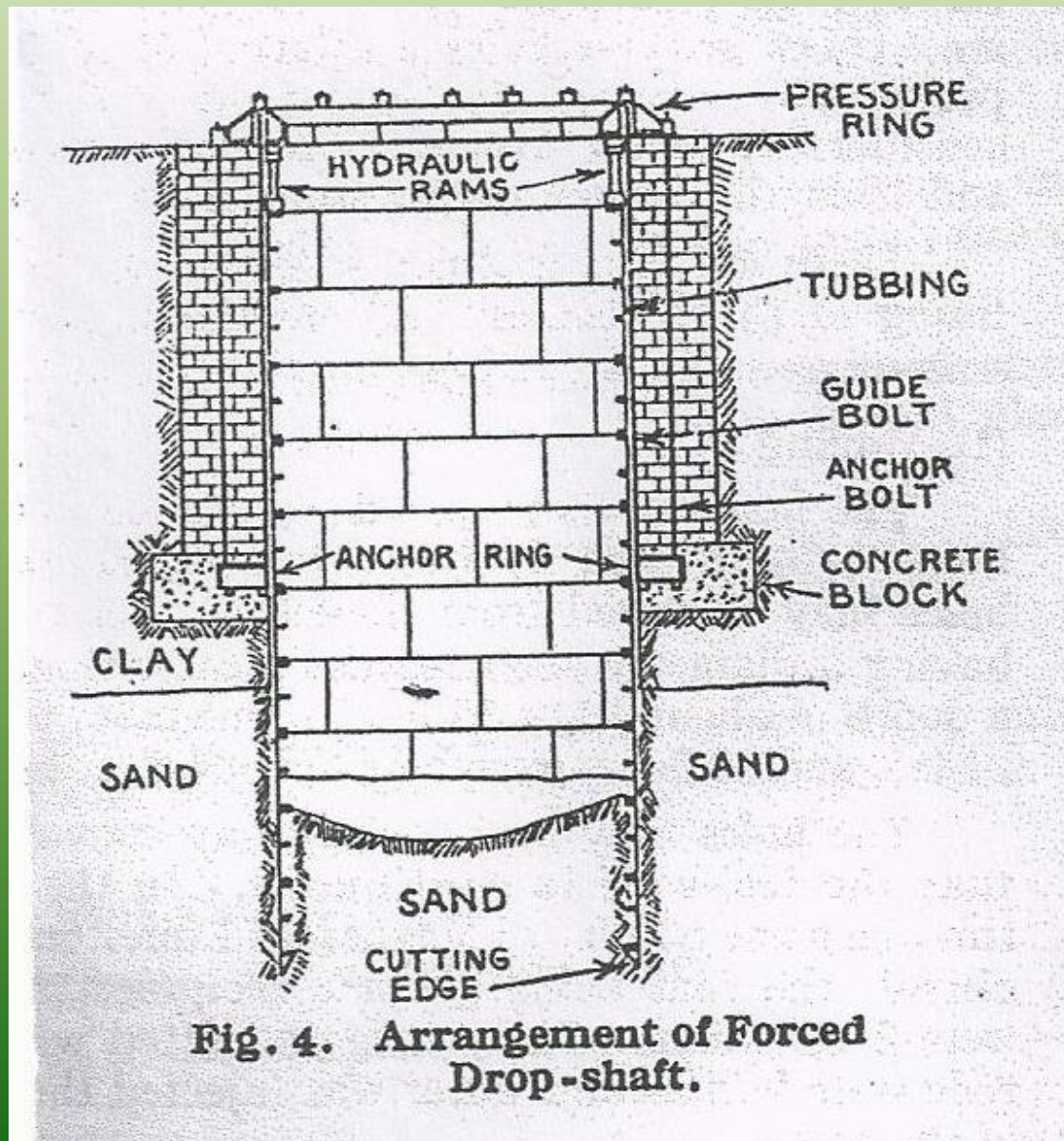
Sinking drum process

- Disadvantages

- Sometimes difficult to keep the drum vertical
- Skin friction increases rapidly with depth
- Danger of drum being sticking altogether
- In running ground, large amount of excess material is often excavated, resulting in subsidence of surrounding surface and damaging adjacent structures

It is not advisable to install permanent headgear and winding engines till sinking through loose ground completed

Forced drop shaft method



Forced drop shaft method

- Applicable to cases where beds are known to consist of alternate tough and loose ground in which ordinary sinking drum will either refuse to descend or would not sink far enough to prevent excessive quantities of loose material entering the shaft from below cutting shoe

Forced drop shaft method

- It consists of jacking or forcing down by hydraulic rams one (or more) cast iron drums of internally flanged tubing within a preliminary caisson of brick work or concrete
- The brick work caisson may form the walling in the upper part of the shaft
- The hydraulic rams re-act against a massive cast iron pressure ring erected on top of brick work caisson, connected by number of stout vertical anchor bolts and guide bolts to a strong anchor ring

Forced drop shaft method

- Anchor bolts are embedded in the brick work
- Guide bolts are close to the inner periphery to form guide for tubbing
- Pressure and anchor rings are strong to resist enormous pressure of rams
- When the tubbing has been forced down, another ring is added to the top

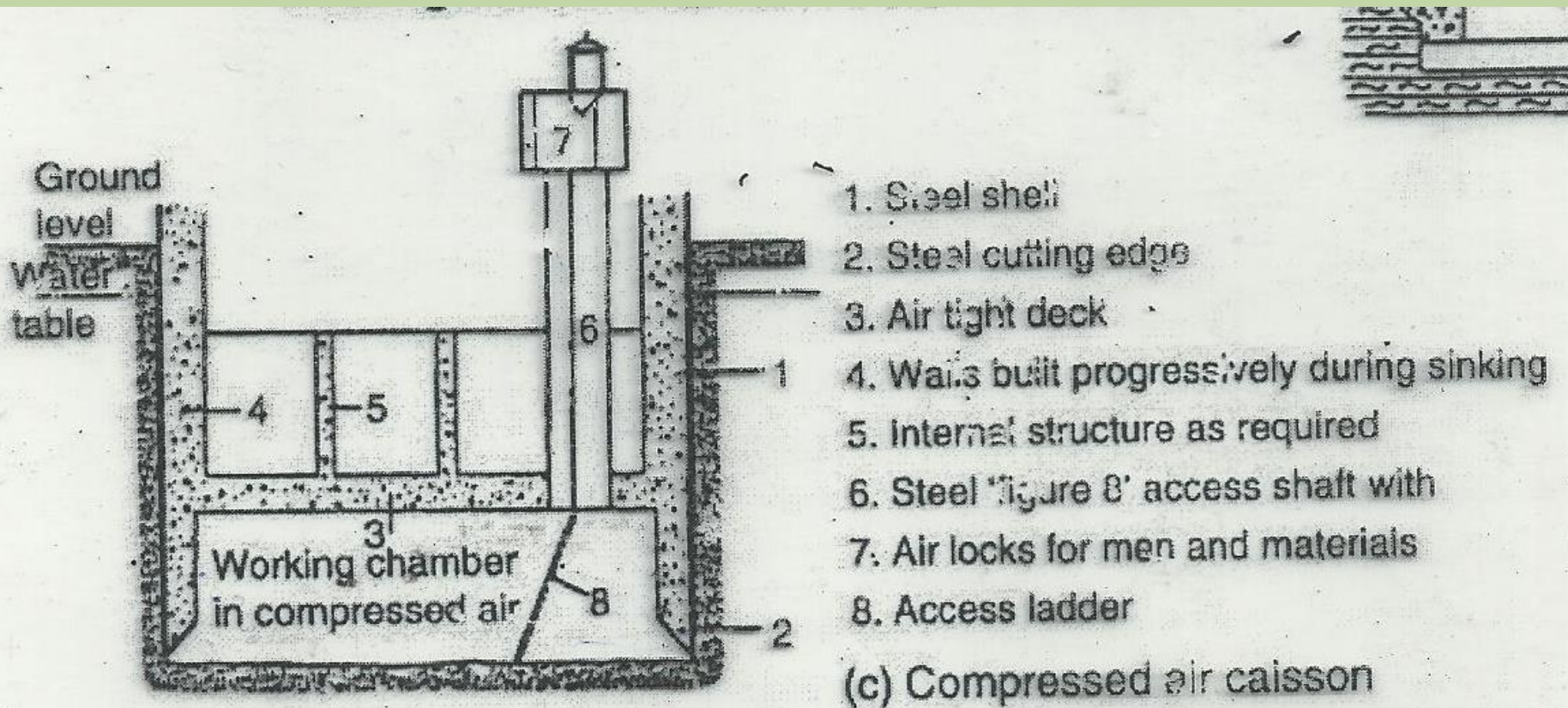
Forced drop shaft method

- Advantages
 - Much greater depth of loose ground can be pierced
 - Diameter of excavation is not excessive
 - More certain method

Pneumatic caisson method

- Designed for waterlogged quick sand or mud
- Lower portion of the drum is converted virtually into a diving bell by means of partition or diaphragm, 6 or 7 feet above the shoe
- Compressed air at a pressure exceeding that of the surrounding water is led into the chamber so formed
- An airlock is mounted on top of the chamber to permit passage of men and materials
- The caisson sinks by gravity

Pneumatic caisson method



Pneumatic caisson method

- Disadvantages
 - Working in compressed air is injurious to health – cause caisson sickness
 - Limiting depth is about 100 ft
 - Slow progress and costly
 - Compressed air is liable to be vitiated
 - Relatively higher temp of comp air